# Mustafa Mahmood

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## PERSONAL PROFILE

First Class MEng Mechatronic Engineering graduate (University of Manchester) with 12 months of industrial experience as an Embedded Software Engineer at Renishaw, developing real-time C++ application firmware for CMM metrology controllers. Robotics software engineer with hands-on experience spanning the embedded-to-operator-interface stack: ROS2 development in Python and C++ (nodes, publishers/subscribers, services/clients), physical robot calibration and deployment (laser scan and odometry processing on real hardware), and control software for physical platforms including stepper motor control and sensor/actuator interfacing on a custom ESP32-S3 rig, paired with a live-telemetry Python GUI built for that same platform. Comfortable working across the software/hardware boundary, from diagnosing a race condition on a physical test rig to re-tuning control parameters directly on deployed robot hardware. Early-career engineer bringing rigorous TDD practice, SOLID/GoF design discipline, and a hands-on, testing-focused approach to real-world robotic systems.

## INDUSTRIAL EXPERIENCE

**Embedded Software Engineer (Industrial Placement)** *Sept 2023 – Sept 2024*

*Renishaw plc, New Mills, UK*

- Developed application-layer embedded C++ firmware for Coordinate Measuring Machines (CMMs), high-precision metrology systems used in aerospace, automotive, and manufacturing quality control.
- Built a joystick (Manual Control Unit) emulator over RS485 against a documented communication spec, to observe and validate how the CMM controller reads and writes its registers.
- Wrote 100+ unit tests using TDD/GMock, validating read/write behaviour of controller commands and enabling confident refactoring of hardware-facing logic.
- Applied SOLID principles and Gang of Four design patterns (State, Strategy) plus pub-sub integration, formalised through UML design work used in code reviews.
- Bench-validated firmware behaviour against real CMM controller hardware ahead of formal code review.

## RELEVANT PROJECTS

**ROS2 C++ Development (The Construct)**

- Progressing from the guided Python coursework into C++, working directly with core ROS2 concepts: nodes, publishers, subscribers, timers, services, and clients.
- Built a C++ node subscribing to laser scan, odometry, and mission-command topics, publishing Twist commands to navigate a rover to mission-specified coordinates with obstacle avoidance prioritised over navigation; passed the course's automated grading criteria.
- Developed a separate C++ wall-following node using segmented laser scan data to maintain a target distance from a wall, validated in simulation.
- Currently building a ROS2 service using the request/response client-server pattern, wrapping a course-provided OpenCV image-processing pipeline to return a classification result via the service response.

**ROS2 Robot Control — Python OOP**

- Independently completed the Construct's ROS2 Python course to develop software and robotics skills.
- Built a Python robot control class in ROS2 covering velocity control, timed motion, laser scanner processing, and odometry tracking.
- Wrote a naive obstacle avoidance algorithm using segmented laser scan data across five directional regions to select turn direction.
- Implemented 16-point direction tracking from odometry yaw, and obstacle/wall classification via statistical mode analysis on frontal laser data.
- Calibrated laser index mappings for the real FastBot (450-element array) by physically orienting it toward walls, distinct from the simulated Summit XL (660-element array); independently tuned angular velocity on the physical robot after default values proved too low for reliable turning.
- Validated all algorithms in Gazebo simulation (Summit XL) before deploying to the physical FastBot.

**Firmware-Controlled Tensile Testing Rig (MEng Final Project)**

- Designed a real-time data acquisition system for uniaxial tensile testing using the ESP32-S3 microcontroller.
- Developed a stepper motor control algorithm exploiting the ESP32-S3's hardware peripherals for accurate, jitter-free motor step timing independent of CPU load.
- Interfaced a load cell and 24-bit load cell amplifier for force measurement.
- Decoded quadrature pulses from a glass linear scale via the ESP32-S3 hardware pulse counting peripheral for strain measurement with a ±10 µm accuracy.
- Designed and iteratively refined a Python GUI with live telemetry, six real-time graphs and CSV export, based on supervisor and end-user feedback.
- Investigated and documented a race-condition bug between an in-flight motion pulse and a stop command, identified approximately two weeks before the project deadline, and produced a detailed handover write-up of the issue and fix.

**Mobile Robots & Autonomous Systems — MATLAB/Simulink Coursework (MEng)**

- Completed the differential-drive robot kinematic model in MATLAB (constraint matrices, forward kinematics) and extended it into Simulink for time-domain simulation, validating output against hand-calculated theory.
- Implemented the core Kalman filter predict/update equations from lecture notes; used provided fusion scripts to combine gyroscope and compass/IMU data, tuning process/measurement noise covariances (Q, R) to observe filter behaviour.
- Wrote code to reconstruct a robot's 2D path from real 6-axis IMU data using trapezoidal integration, observing the effect of drift from sensor noise.
- Used provided MATLAB image-processing functions (thresholding, mean/Gaussian filtering, Sobel/Canny edge detection) to observe the effect of varying kernel size and threshold parameters.
- Generated grid/graph representations of maps and used provided search algorithm implementations (BFS, Dijkstra, Wavefront, A*) to compare shortest paths and routes.

## TECHNICAL SKILLS

**Programming & Practice:** Python, C, Modern C++ (OOP, SOLID, GoF patterns), MATLAB/Simulink; TDD/GMock unit testing, integration testing, Git, UML design
**Robotics, ROS2 & Control:** ROS2 in C++ and Python (nodes, publishers/subscribers, services/clients, launch files); laser scan and odometry-based navigation, Gazebo simulation, physical robot deployment; sensor fusion (Kalman filtering), path planning (A*, Dijkstra, BFS, Wavefront), occupancy/topological mapping fundamentals
**Embedded & Sensor Integration:** STM32, ESP32-S3, RS485/UART protocols, quadrature decoding, ADC/load cell interfacing
**Other:** Linux, Bash scripting; English (fluent), Urdu (fluent), Spanish (basic)

## EDUCATION

**MEng Mechatronic Engineering (Industrial Route)** *2021 – July 2026*

*The University of Manchester, Manchester, UK — First Class Honours*

- Electronics Project module: diagnosed and documented 4 faults on a PCB against schematic and placement diagrams, using continuity testing, polarity checks, and multimeter measurement to isolate faulty tracks and components.
- Relevant modules: Robotic Manipulators (85), Control Systems II (90), Mobile Robots & Autonomous Systems (77), System Identification & Artificial Intelligence (83), Digital Signal Processing (85).

**A-Levels (Loreto Sixth Form College)** *A* A* A*

## ADDITIONAL INFORMATION

British Citizen | Full UK Driving Licence | Willing to relocate